Network analysis of **wildfire** transmission and implications for risk governance

AA Ager, CR Evers, MA Day, HK Preisler, AMG Barros… - PLoS …, 2017 - journals.plos.org

… to **map** and predict future patterns of potential **wildfire** transmission at … by newer US federal **wildfire management** policy that … in terms of ignition probability and **wildfire spread** rates [26]. …

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A brief **review** of machine learning algorithms in forest fires science

R Alkhatib, W Sahwan, A Alkhatieb, B Schütt - Applied Sciences, 2023 - mdpi.com

… **forest fire prediction** in several vital areas of **forest fire** … models to predict the **spread** and **behavior** of forest fires using a … were developed to **map forest fire susceptibility** in the northern …

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A **review** on early **forest fire** detection systems using optical remote sensing

P Barmpoutis, P Papaioannou, K Dimitropoulos… - Sensors, 2020 - mdpi.com

… aiming to support **wildfire management** at the national and regional level use either satellites or combine them with a small fleet of planes to detect and **map** the extent, **spread**, and …

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Provision of a **wildfire** risk **map**: informing residents in the wildland urban interface

P Mozumder, R Helton… - Risk Analysis: An …, 2009 - Wiley Online Library

… to promote **wildfire** risk mitigation **behavior** in WUI zones. … for the **wildfire** risk information (our focus here), the **survey** … As a check on robustness, we report multiple **model** specifications …

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Performance evaluation of machine learning methods for **forest fire** modeling and **prediction**

BT Pham, A Jaafari, M Avand, N Al-Ansari, T Dinh Du… - Symmetry, 2020 - mdpi.com

… This study and the resultant **susceptibility maps** provide a … -**survey** methods for the **prediction** of **forest fire susceptibility** by … /method has been yet identified to capture fire **behavior** in all …

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Forest fire and degradation assessment using satellite remote sensing and geographic information system

PS Roy - Satellite Remote sensing and GIS applications in …, 2003 - unisdr.org

… wide wild **spread** of the **forest fire**. Figure-1 shows triangle of … The **forest fire management** system in Thailand has its strong … also to prepare **forest fire** risk area/disaster **map** for GIR forest …

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Analysis of evacuation **behavior** in a **wildfire** event

T Toledo, I Marom, E Grimberg, S Bekhor - International journal of disaster …, 2018 - Elsevier

… place, is critical for the efficient **management** of the event. The usual … A large-scale **wildfire** took place in Haifa on November 24, … Shortly after the fire events, a web **survey** was developed …

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Review of users' needs in operational fire danger estimation: the Oklahoma example

JD Carlson, RE Burgan - International Journal of remote sensing, 2003 - Taylor & Francis

… to be useful to the **wildland fire management** community in … For example, the Canadian **Forest Fire Behavior Prediction** … **maps** for the various components of the empirical **Forest Fire** …

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Satellite remote sensing contributions to **wildland fire** science and **management**

E Chuvieco, I Aguado, J Salas, M García… - Current Forestry …, 2020 - Springer

… We **review** in this paper the main research trends on the use … their seasonal outlook for fire and **map** at-risk areas as well … to the main fire **propagation** media. Different classification …

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Mapped versus actual burned area within **wildfire** perimeters: characterizing the unburned

CA Kolden, JA Lutz, CH Key, JT Kane… - … and **Management**, 2012 - Elsevier

… **wildfire** studies have utilized fire occurrence as the primary data source for investigating the causes and effects of **wildfire** … , and investigated patch **dynamics** of unburned patches within …

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Wildland fire potential: A tool for assessing **wildfire** risk and fuels **management** needs

G Dillon, J Menakis, F Fay - In: Keane, Robert E.; Jolly, Matt; …, 2015 - research.fs.usda.gov

… torching, crowning, and other forms of extreme **wildfire behavior** under conducive weather conditions. We chose to produce the WFP **map** for all lands in the conterminous United States (…

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Surging **wildfire** activity in a grassland biome

VM Donovan, CL Wonkka… - Geophysical Research …, 2017 - Wiley Online Library

… how **wildfire dynamics** are changing in this grassland biome. … level of the hurdle **model**: **wildfire** number and total area … of our hurdle models were **mapped** by Great Plains L3 ecoregion. …

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Framework for an Australian fuel classification to support **bushfire management**

JJ Hollis, JS Gould, MG Cruz… - Australian …, 2015 - Taylor & Francis

… fire authorities, and a **review** of other classification systems used … concept focus on the variables known to drive surface fire **spread** … simulations at smaller spatial scales if no **mapping** or …

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Fuel models and fire potential from satellite and surface observations

RE Burgan, RW Klaver, JM Klaver - … journal of **wildland fire**, 1998 - connectsci.au

… data, then refined through **review** by fire managers familiar … of a national 1 km2 fuel **model map** for the United States and … probability that a **wildland fire** will ignite and **spread** is strongly …

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A comparison of three approaches for simulating fine-scale surface winds in support of **wildland fire management**. Part I. **Model** formulation and comparison against …

JM Forthofer, BW Butler… - … Journal of **Wildland Fire**, 2014 - CSIRO Publishing

… the mass-conserving **model** to ~1 h for the momentum-conserving **model** using consumer-… -time fire **spread prediction** models to support fire **management** activities, **mapping** local wind …

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The acceptance of using information technology for disaster risk **management**: A systematic **review**

K Meechang, N Leelawat, J Tang, A Kodaka… - Engineering …, 2020 - engj.org

… For example, fire **map** on web platform showed **prediction** and visualization of … fire **prediction** and fire **behavior** visualization [84]. The interface was developed for predicting **wildfire** and …

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GIS-grid-based and multi-criteria analysis for identifying and **mapping** peat swamp **forest fire** hazard in Pahang, Malaysia

I Setiawan, AR Mahmud, S Mansor… - … and **Management**: An …, 2004 - emerald.com

… develop a **forest fire** hazard **model** and **map** in order to identify, classify and **map** fire hazard … Topographic factors which influence the **spread** of fire are incorporated with the proximity of …

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FireStation—an integrated software system for the numerical **simulation** of fire **spread** on complex topography

AMG Lopes, MG Cruz, DX Viegas - Environmental Modelling & Software, 2002 - Elsevier

… The **prediction** of **forest fire behavior** is an essential … Table 1, based on the **review** by these authors, synthesizes relevant … The main view depicts the fuel **map** with the simulated fire …

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Mapping wildland fuels for fire **management** across multiple scales: Integrating remote sensing, GIS, and biophysical modeling

RE Keane, R Burgan… - … Journal of **Wildland Fire**, 2001 - connectsci.au

… to the **prediction** of fire **behavior** because they do … **mapping** fuels for fire **management** at multiple scales. We discuss challenges involved in **mapping** fuels, **review** historical fuel **mapping** …

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Inundation, flow **dynamics**, and damage in the 9 January 2018 Montecito debris-flow event, California, USA: Opportunities and challenges for post-**wildfire** risk …

JW Kean, DM Staley, JT Lancaster… - …, 2019 - pubs.geoscienceworld.org

… Our **mapping** follows the five flow paths that sustained the most damage: Montecito (including the Cold … By linking debris-flow impacts to the established Hazus **model**, our results can be …

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